RAHUL MITRA

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EDUCATION

National University of Singapore **Aug 2023 – Jul 2027**
Bachelor of Engineering, Industrial and Systems Engineering (GPA: 4.37/5.00)

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.

WORK EXPERIENCE

Warehouse AI & Operations Research Engineer (Part-Time), STMicroelectronics (NUS Capstone) **Aug 2026 – Present**

- Architected a layout-agnostic AI put-away engine reading each site's shelf coordinates, dimensions, and operating rules, so one model fits any STMicroelectronics warehouse without per-site rework.

Global Operational AI Systems & Data Engineer (Part-Time), Abbott Laboratories (Contract) **Jun 2026 – Present**

- Developed ADAM, an offline tool-calling LLM assistant (Gemma 4 E4B on llama.cpp) for 6 spray dryers across 4 global sites, giving engineers self-serve answers to decisions previously requiring process-control specialists.
- Implemented a Seeq signal-analytics toolkit (Kalman smoothing, spectral and lag analysis, change-point detection, matrix profiles) and a 5-stage offline handbook RAG pipeline (OCR, ONNX embeddings, Chroma, LangChain).

Data Analytics & Product Management Intern, Abbott Laboratories (Converted to Contract) **Jan 2026 – Jun 2026**

- Engineered a SimPy discrete-event digital twin and OR-Tools CP-SAT solver for hybrid flow-shop scheduling, comparing 3 approaches (heuristic, MIP, genetic algorithm) to produce decision-ready schedules.
- Trained regional engineering teams across Asia and Europe on repeatable AI workflows spanning 7 areas: prompting, model selection, benchmarks, cost awareness, agents, validation, and safeguards.

Software Engineer (Full Stack & Cloud), People's Association (Pro bono) **Sep 2025 – Sep 2026**

- Designed, built, and deployed Churp, a resident-driven garden-plot balloting platform now live in Tengah and slated for national rollout, seeded by a S$20K Sparks Community Innovation Fund award.

RESEARCH EXPERIENCE

Computer Vision Researcher, BlendED x Amazon Robotics (AI+X Global Challenge) **Jul 2026 – Oct 2026**

- Built vision systems turning raw camera feeds into perception-ready frames across 3 workstreams: camera ISP, super-resolution, and image restoration under real-world blur.
- Reproduced a published UNet super-resolution model in PyTorch from broken author code, recovering its undocumented evaluation protocol and training 12 models, matching reported results at 8x and landing within 0.78 dB at 2x and 4x.

PROJECTS

Team Lead & Finalist, Swarmline: SG Defence Tech Hackathon (1,300+ applicants) **Sep 2026**

- Directed a 5-person team to architect decentralised drone-swarm coordination on correlated random-walk search, gradient-flow dispersion, delay-tolerant gossip, cross-inhibition consensus, and response-threshold task allocation.

Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams) **Jan 2026**

- Built an autonomous loop managing electricity and utility savings across multiple HDB households at once on SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.

Team Lead, Maritime Deficiency Severity Forecasting: Maritime Hackathon **Jun 2025**

- Led model training for a maritime-inspection NLP system, fine-tuning a 109M-parameter BERT with DNN layers and gradient clipping on the ASPIRE 2A supercomputer.

CO-CURRICULAR ACTIVITIES AND VOLUNTEER EXPERIENCES

Elderly Care Volunteer, NTUC Health Active Ageing Centre **Sep 2025 – Sep 2026**

- Coordinated weekly senior-engagement activities, including home visits to check on residents' wellbeing, hear life stories, and help with everyday tasks, building empathy for accessible, community-focused product design.

SKILLS AND CERTIFICATIONS

Technical: Python, PyTorch, BERT, DNN, RAG, AI agents, GPT-4, Gemini, Claude Code, ChatGPT, Codex, ComfyUI, SEALION, ONNX, Chroma, local LLM serving (llama.cpp), vector search, model evaluation, NLP; vision: camera ISP, super-resolution, image restoration, camera models and calibration, fundamental and essential matrices, epipolar geometry, absolute pose estimation, three-view geometry, structure-from-motion (SfM) with bundle adjustment, two- and multi-view stereo, generalised camera models; HPC: ASPIRE 2A, CUDA; deep RL (PPO, A2C, DDPG, DQN)
Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery